



Butterflyfish (Chaetodontidae) Species in Three Small Outer Islands of Papua, Indonesia (Liki, Bepondi and Miossu)

Risandi Dwirama Putra^{1,2*}, Petrus Christianus Makatipu³, Beginer Subhan⁴,
Dietrich Geoffrey Bengen⁴, Neviaty Putri Zamani⁴, Muhammad Abrar³,
Rikoh Manogar Siringoringo³, Frensy Damianus Hukom³, Ludi Parwadani Aji^{3,5},
Efrin A. Dollu¹, Asniati Ningsi¹, and T. Ersti Yulika Sari⁶

¹Doctoral Program of Marine Science, Institut Pertanian Bogor, Indonesia.

²Department of Naval Architecture, Universitas Maritim Raja Ali Haji, Indonesia.

³Research Center for Biota Systems, National Research and Innovation Agency, Indonesia.

⁴Department of Marine Sciences and Technology, Institut Pertanian Bogor, Indonesia.

⁵Marine Evolution and Ecology Group, Naturalis Biodiversity Center, The Netherlands.

⁶Faculty of Fisheries and Marine Sciences, Riau University, Indonesia.

*Corresponding Author: risandi1989dwirama@apps.ipb.ac.id

ARTICLE INFO

Article History:

Received: Feb. 10, 2026

Accepted: May 19, 2026

Online: June 9, 2026

Keywords:

Coral,
Diversity,
Expedition,
Reef fishes,
Outer Islands

ABSTRACT

The Liki, Bepondi, and Miossu islands are remote outer islands of Indonesia located in West Papua, near the equator within the Coral Triangle, a global hotspot of marine biodiversity. Despite their ecological significance and unique oceanographic exposure to the western equatorial Pacific, baseline data on reef fish assemblages remain critically limited. This study presents the first systematic assessment of butterflyfish (Chaetodontidae) diversity, a key bioindicator group, across these islands. Findings reveal spatial heterogeneity in species composition, underscoring the biogeographic distinctiveness of Indonesia's outer reefs. These insights are essential for informing evidence-based conservation strategies and advancing biodiversity monitoring in the eastern Indonesian seascape. Butterflyfish surveys were conducted using the standardized Underwater Visual Census (UVC) technique along linear belt transects measuring 70 meters in length, with an effective observation width of 2.5 meters on each side, encompassing a total survey area of 350 m² per transect. The census focused on quantifying species richness and relative abundance of butterflyfish, a taxonomic group widely regarded as an ecological indicator of coral reef condition. A total of 13 reef sites were surveyed across three island groups: six at Liki Island, three at Bepondi Island, and four at the Miossu Islands. The survey identified 30 species of butterflyfish across four genera during an expedition surrounding Liki, Bepondi, and the Miossu Islands. The Kruskal–Wallis test identified highly significant differences in butterflyfish abundance across the three outer island groups ($\chi^2 = 35.69$, $p = 1.77 \times 10^{-8}$), accompanied by a large effect size ($\epsilon^2_{\text{ordinal}} = 0.40$), indicating substantial variation among sites. Subsequent pairwise comparisons using Dunn's test with Bonferroni adjustment confirmed that butterflyfish abundance at Liki Island and the Miossu Islands was significantly higher than that observed at Bepondi Island ($p = 5.58 \times 10^{-8}$).

INTRODUCTION

The Liki, Bepondi, and Miossu are among Indonesia's outer islands, located in West Papua near the equator within the heart of the Coral Triangle area. Their geographic position plays a crucial role in shaping marine biodiversity, which varies across different biogeographic scales (**Mumby, 2001**) and is distributed across the region (**Jost *et al.*, 2010**). Previous studies (**Cowman, 2014**) have highlighted the influence of longitudinal and latitudinal gradients on marine biodiversity across various taxonomic groups, with shifts in species composition and diversity along these gradients (**Johnson *et al.*, 2019**).

Marine latitudinal diversity gradients are widely recognized as key indicators of biodiversity patterns, particularly for reef fish (**Brandl *et al.*, 2018; De Brauwer *et al.*, 2019**). The richness of reef fish species tends to increase toward lower latitudes (**Hoeksema, 2007; Jones, 2013; Chabanet *et al.*, 2016**), which also impacts their abundance and biomass (**Johnson *et al.*, 2019**). Additionally, environmental factors such as wave-generated turbulence (**Choat & Bellwood, 1991**), temperature fluctuations (**Cole, 2001; Robertson *et al.*, 2004**), and ocean acidification (**Hughes *et al.*, 2003; Hoegh-Guldberg *et al.*, 2007; Maduppa *et al.*, 2014; Elias, 2017**) contribute to biodiversity distribution by influencing coral reef stability and habitat availability (**Fabrizius *et al.*, 2005**). These conditions ultimately lead to the highest levels of marine biodiversity occurring in lower tropical latitudes (**Allen, 2000**).

The small outer islands of West Papua possess unique oceanographic characteristics due to their direct exposure to the western equatorial Pacific Ocean. They experience deep waters, clear visibility, dynamic wave action, and strong currents (**Surinati & Corvianawatie, 2019**). The region is also influenced by the Indo-Pacific Warm Pool, known for its elevated sea surface temperatures (SST), which significantly correlate with species richness and promote biodiversity and species evolution (**Sanciangco *et al.*, 2013; Nicol *et al.*, 2014**). It has been identified that the Coral Triangle's warm waters is a key factor in supporting the world's most diverse coral reef ecosystems (**Förderer *et al.*, 2018**). Despite the ecological significance of Liki, Miossu, and Bepondi islands, they exhibit contrasting geomorphological and oceanographic features that appear to differentially shape coral reef ecosystem structure and associated biodiversity. Liki and Miossu are both classified as high islands, typified by steep coastal topography, narrow fringing reefs, and deeper reef slopes, which likely enhance habitat heterogeneity and water exchange. In contrast, Bepondi Island is a low-lying carbonate platform dominated by broad reef flats with reduced vertical complexity. These geomorphic disparities exert significant influence on local hydrodynamics, sedimentation patterns, and benthic habitat structure factors that are known to modulate species richness and community composition in reef-associated faunal assemblages. The observed variation in Chaetodontidae diversity and abundance among the islands likely reflects these underlying physical and ecological gradients.

This study aims to systematically investigate the diversity and spatial distribution patterns of butterflyfish (Chaetodontidae) across three geomorphologically distinct outer

islands in West Papua. As a bioindicator family intimately linked to coral reef condition, Chaetodontidae is a valuable proxy for assessing reef health and ecological integrity. By establishing a robust baseline dataset in this underexplored region of the Coral Triangle, the study contributes essential empirical evidence to support adaptive marine spatial planning, biodiversity monitoring, and targeted conservation interventions in Indonesia's peripheral reef ecosystems.

MATERIALS AND METHODS

1. Research locations

This study was conducted across 13 coral reef sites distributed among three outer islands of West Papua: six stations on Liki Island, three on Bepondi Island, and four on Miossu Island. The allocation of sites was proportionally adjusted based on the relative spatial extent of each island's reef system, with Liki Island requiring a greater number of sampling sites due to its larger area, in order to ensure adequate ecological representation. Field data were collected during the Nusa Manggala scientific expedition, conducted over an extended survey period from October to December 2018. The precise geographic coordinates and spatial configuration of the sampling stations are presented in Fig. (1) and detailed in Table (1).

Table 1. Information on coral reef fish data collection and GPS coordination of Liki, Bepondi, and Miossu islands.

Site Name	Location	Latitude	Longitude	Topography
LIKC.02	Liki	S: 1° 37' 34.255"	E: 138° 43' 43.411"	Reef Slope
LIKC.04	Liki	S: 1° 38' 43.451"	E: 138° 46' 47.402"	Reef Slope
LIKC.06	Liki	S: 1° 37' 42.305"	E: 138° 44' 42.863"	Reef Slope
LIKC.09	Liki	S: 1° 36' 52.902"	E: 138° 42' 49.053"	Reef Slope
LIKC.21	Liki	S: 1° 35' 32.028"	E: 138° 43' 56.024"	Reef Slope
LIKC.26	Liki	S: 1° 37' 12.299"	E: 138° 44' 54.211"	Reef Slope
BEPC.01	Bepondi	S: 0° 23' 31.708"	E: 135° 16' 03.154"	Reef Flat
BEPC.04	Bepondi	S: 0° 24' 29.822"	E: 135° 15' 19.619"	Reef Flat
BEPC.07	Bepondi	S: 0° 24' 28.202"	E: 135° 16' 28.462"	Reef Flat
MIOC.02	Miossu	S: 0° 20' 41.399"	E: 135° 09' 06.497"	Reef Slope
MIOC.12	Miossu	S: 0° 20' 47.641"	E: 132° 10' 12.867"	Reef Slope
MIOC.25	Miossu	S: 0° 21' 33.304"	E: 132° 11' 17.250"	Reef Slope
MIOC.34	Miossu	S: 0° 22' 00.815"	E: 132° 11' 41.643"	Reef Slope

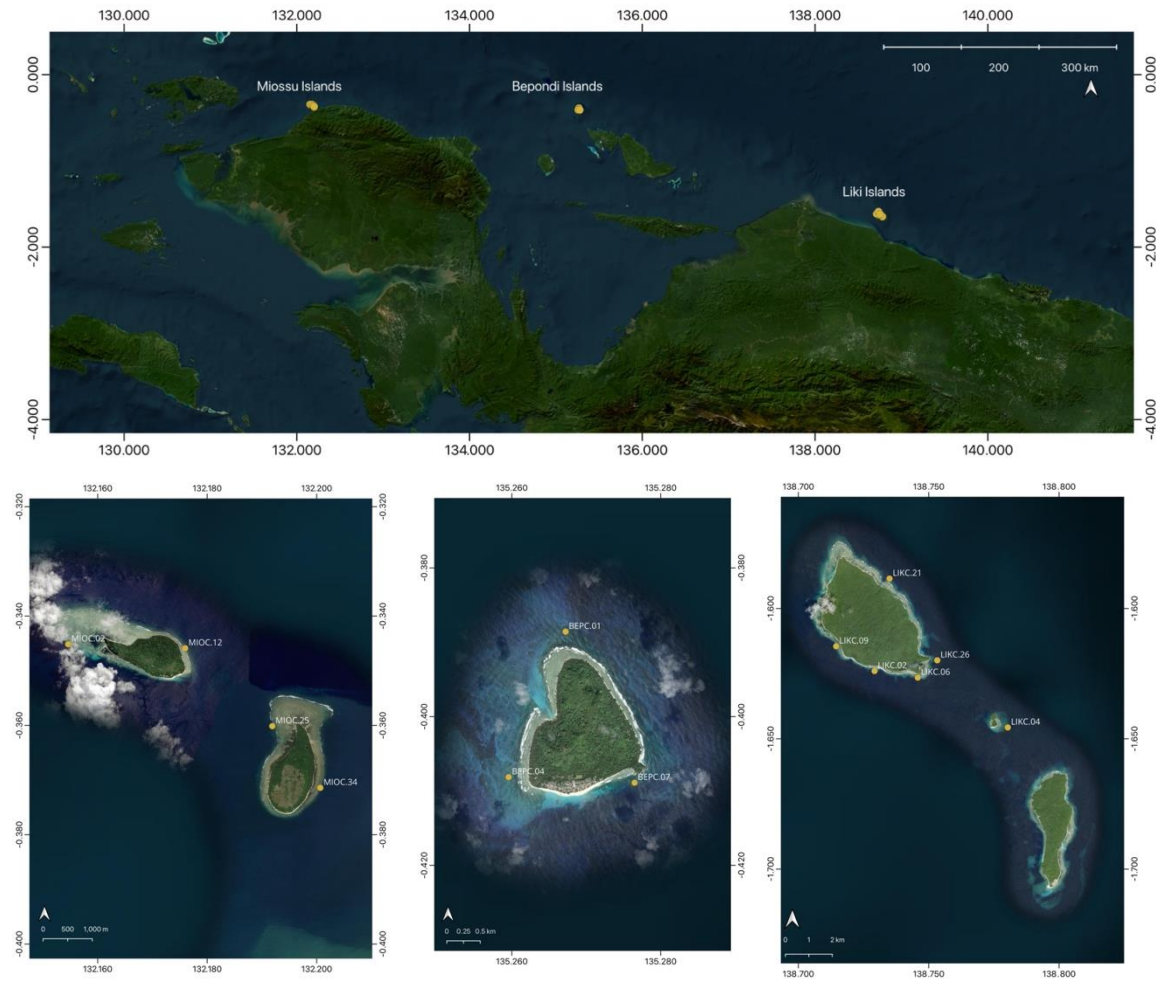


Fig. 1. The research sites were distributed across three island locations: the Miossu Islands (stations MIOC.02, MIOC.12, MIOC.25, MIOC.34), Bepondi Island (stations BEPC.01, BEPC.04, BEPC.07), and Liki Island (stations LIKC.02, LIKC.04, LIKC.06, LIKC.09, LIKC.21, LIKC.26).

2. Chaetodontidae Data Collection

The identification of reef fish species in Liki Island, Bepondi Island, and the Miossu Islands was conducted using the Underwater Visual Census (UVC) method (**Buxton & Smale, 2006; Floros *et al.*, 2013; Corrales *et al.*, 2015; Fatimah *et al.*, 2018; Putra *et al.*, 2018; Rovira *et al.*, 2019**). This method was adapted from **English *et al.* (1997)**. For this study, butterflyfish (Chaetodontidae) abundance and species composition were recorded using standardized belt transect surveys, each measuring 70 meters in length and 5 meters in width, covering an effective survey area of 350 m² per transect. At each sampling site, a transect tape was deployed along a depth contour between 5 and 10 meters and positioned parallel to the shoreline to ensure consistency in habitat type and depth gradient (**Giyanto *et al.*, 2014**). The UVC methodology is widely employed in coral reef ecosystem assessments to determine the diversity and abundance of reef fishes,

macroalgae, and benthic invertebrates (Jones *et al.*, 2015). Additionally, by recording the total length of observed fish, this method allows for the estimation of reef fish biomass (Friedlander & DeMartini, 2002; Cresson *et al.*, 2019). Fish species data were collected at each research site through underwater diving sessions lasting between 50 and 80 minutes (Sandin *et al.*, 2008; Corrales *et al.*, 2015; Chabanet *et al.*, 2016). Observed fish were recorded using a slate and subsequently re-identified through underwater photographs to ensure a more accurate taxonomic classification. Species identification was performed based on reference materials, including Kuiter and Tonozuka (2001a, b, c), Allen *et al.* (2003), Du *et al.* (2016), and FishBase (Froese & Pauly, 2020).

3. Data Analysis

In this study, a detailed account was provided of the coral reef fish species recorded during research on Liki Island, Bepondi Island, and the Miossu Islands. The findings are presented in a table listing the family, genus, and species of the identified reef fish. Additionally, a non-parametric statistical analysis was conducted using the Kruskal-Wallis test to compare coral reef fish density across the three islands. Furthermore, fish species classified as Near Threatened or higher in conservation status were identified and reported, along with the number of individuals recorded during the surveys. The abundance of Chaetodontidae fish at each sampling station was also analyzed by comparing the conditions across the islands of Liki, Bepondi, and Miossu using the Kruskal-Wallis test. A comprehensive correlation matrix analysis was conducted to determine potential similarities in Chaetodontidae communities among the three islands. To further validate the similarities in Chaetodontidae conditions, we performed a cluster dendrogram analysis, which provides a detailed hierarchical representation of the similarity in species composition and abundance across sampling sites on Bepondi Island (BEPC), the Miossu Islands (MIOC), and Liki Island (LIKC). This multi-faceted approach offers a robust understanding of the spatial distribution and ecological patterns of Chaetodontidae populations in these regions.

RESULTS

1. The List of Butterflyfish (Chaetodontidae) Species and Their Abundance in Liki, Bepondi and Miossu Islands

A total of 30 butterflyfish species (*Chaetodontidae*) were recorded across the 13 reef stations distributed among the Liki, Bepondi, and Miossu Islands (Table 2). Butterflyfish abundance was expressed as mean individuals per site $\pm$ standard deviation (SD), enabling comparisons of both central tendency and variation across island groups. Among the surveyed islands, Liki Island exhibited the highest overall mean abundance and species richness of *Chaetodontidae*, with several taxa showing notably high occurrence, including

Chaetodon kleinii (15.17 ± 11.44), *Hemitaurichthys polylepis* (14.33 ± 21.78), *Heniochus varius* (6.50 ± 2.74), and *Chaetodon ornatissimus* (6.17 ± 3.37). The Miossu Islands similarly supported a high diversity of butterflyfish, notably *C. kleinii* (10.75 ± 2.50), *C. ornatissimus* (8.50 ± 4.65), and *C. lunulatus* (8.00 ± 4.24). In contrast, Bepondi Island consistently demonstrated lower species abundances and limited presence of several key taxa, with many species entirely absent or occurring in low densities (e.g., *C. citrinellus* at 6.67 ± 1.53 being one of few exceptions). Several species showed restricted distribution patterns, including *Hemitaurichthys polylepis*, which was abundant at Liki (14.33 ± 21.78) but completely absent at Bepondi. Indicator species such as *Chaetodon trifascialis* (IUCN Near Threatened) were present across all sites but in low to moderate densities, with the highest occurrence at Liki (2.33 ± 2.25). Interestingly, species categorized as Data Deficient by the IUCN, such as *C. bennetti* and *C. ocellicaudus*, were detected with varying abundances, suggesting the potential importance of these islands for rare or poorly known taxa. The overall spatial pattern indicates clear compositional differences in Chaetodontidae assemblages, with Liki and Miossu supporting more structurally complex and species-rich communities than Bepondi. This variation is likely influenced by geomorphological and oceanographic differences among the islands, including reef slope availability and habitat heterogeneity.

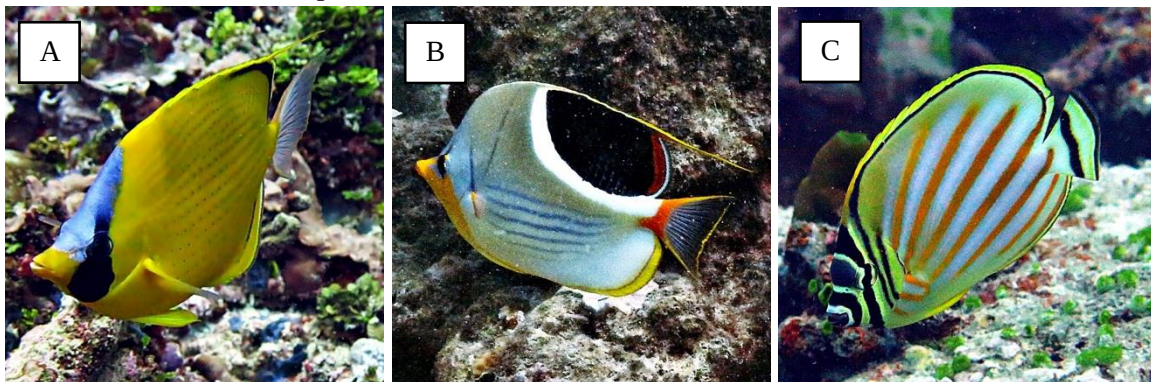
Table 2. List of Chaetodontidae Species and Abundance (mean $\pm$ standard deviation) in Smalls Outer Islands Papua (Liki, Bepondi, and Miossu)

Species	IUCN	Liki n=6 sites	Bepondi n=3 sites	Miossu n=4 sites
<i>Chaetodon adiergastos</i>	LC	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
<i>Chaetodon auriga</i>	LC	0.67 ± 1.03	0.00 ± 0.00	0.75 ± 1.50
<i>Chaetodon baronessa</i>	LC	4.00 ± 1.41	1.33 ± 2.31	4.50 ± 1.73
<i>Chaetodon bennetti</i>	DD	0.67 ± 1.03	3.00 ± 3.61	2.50 ± 1.00
<i>Chaetodon citrinellus</i>	LC	1.67 ± 1.37	6.67 ± 1.53	2.75 ± 0.96
<i>Chaetodon ephippium</i>	LC	4.67 ± 3.93	0.00 ± 0.00	2.25 ± 0.50
<i>Chaetodon kleinii</i>	LC	15.17 ± 11.44	15.00 ± 2.65	10.75 ± 2.50
<i>Chaetodon lineolatus</i>	LC	0.50 ± 1.22	0.00 ± 0.00	2.75 ± 3.50
<i>Chaetodon lunula</i>	LC	0.67 ± 1.63	0.00 ± 0.00	1.25 ± 1.50
<i>Chaetodon lunulatus</i>	LC	7.17 ± 3.31	4.67 ± 3.06	8.00 ± 4.24
<i>Chaetodon melannotus</i>	LC	3.00 ± 2.68	0.00 ± 0.00	4.75 ± 3.69
<i>Chaetodon meyeri</i>	LC	2.17 ± 2.79	1.00 ± 1.73	1.50 ± 1.29
<i>Chaetodon ocellicaudus</i>	DD	2.00 ± 0.63	0.00 ± 0.00	2.50 ± 3.00
<i>Chaetodon ornatissimus</i>	LC	6.17 ± 3.37	3.33 ± 3.06	8.50 ± 4.65

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<i>Chaetodon oxycephalus</i>	LC	3.00 ± 2.53	0.00 ± 0.00	3.75 ± 2.99
<i>Chaetodon. punctatofasciatus</i>	LC	0.67 ± 1.03	0.00 ± 0.00	1.25 ± 2.50
<i>Chaetodon rafflesii</i>	LC	3.00 ± 2.37	0.00 ± 0.00	3.50 ± 1.00
<i>Chaetodon semeion</i>	LC	3.17 ± 1.47	1.00 ± 1.73	3.50 ± 1.73
<i>Chaetodon speculum</i>	LC	1.17 ± 2.04	0.00 ± 0.00	4.00 ± 3.16
<i>Chaetodon trifascialis</i>	NT	2.33 ± 2.25	2.00 ± 2.00	0.75 ± 1.50
<i>Chaetodon ulietensis</i>	LC	3.67 ± 3.39	0.67 ± 1.15	6.25 ± 3.50
<i>Chaetodon unimaculatus</i>	LC	1.17 ± 1.33	0.00 ± 0.00	1.00 ± 2.00
<i>Chaetodon vagabundus</i>	LC	5.83 ± 1.94	0.67 ± 1.15	6.00 ± 2.83
<i>Forcipiger flavissimus</i>	LC	4.50 ± 3.99	1.67 ± 2.89	4.25 ± 1.50
<i>Forcipiger longirostris</i>	LC	1.33 ± 2.16	1.00 ± 1.73	1.00 ± 1.15
<i>Hemitaurichthys polylepis</i>	LC	14.33 ± 21.78	0.00 ± 0.00	2.75 ± 4.86
<i>Heniochus acuminatus</i>	LC	0.31 ± 1.63	0.00 ± 0.00	0.00 ± 0.00
<i>Heniochus chrysostomus</i>	LC	2.50 ± 1.52	3.67 ± 2.89	5.00 ± 1.15
<i>Heniochus singularius</i>	LC	0.33 ± 0.82	0.00 ± 0.00	4.00 ± 2.94
<i>Heniochus varius</i>	LC	6.50 ± 2.74	5.67 ± 1.53	6.75 ± 6.90

The results of this study reveal significant variations in the abundance and distribution of Chaetodontidae species across Liki, Bepondi, and Miossu islands, indicating site-specific differences in reef fish communities. Species such as *Chaetodon kleinii* exhibited relatively high abundance across all three islands, while others, like *Chaetodon trifascialis* (Near Threatened), showed restricted distribution with lower densities, particularly in the Miossu Islands. The presence of data-deficient species (*Chaetodon bennetti* and *Chaetodon ocellicaudus*) highlights the need for further ecological assessments to determine their conservation status and population trends. Variability in abundance, as indicated by high standard deviations in species like *Hemitaurichthys polylepis*, suggests heterogeneous habitat conditions or localized environmental factors influencing reef fish distribution. The findings emphasize the importance of continuous monitoring to assess the impacts of environmental changes and anthropogenic pressures on coral reef fish populations in the outer islands of West Papua.



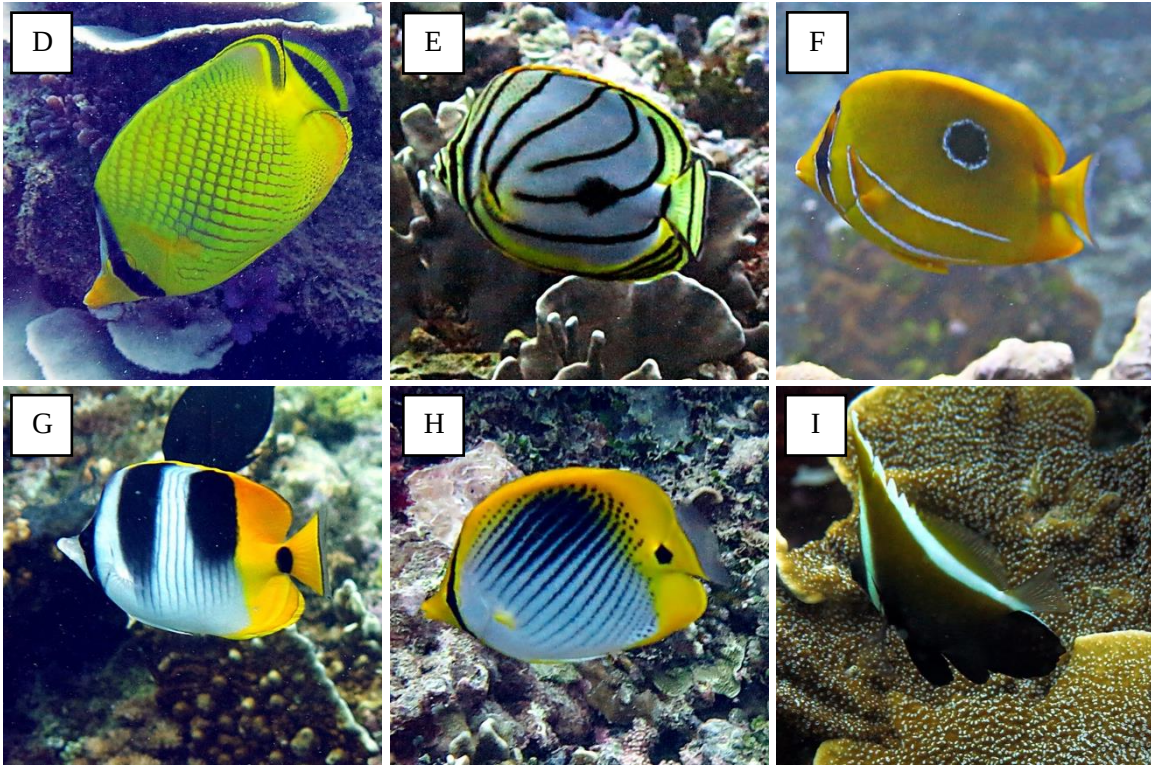


Fig. 2. Direct documentation of several Chaetodontidae species observed across three small outer islands of Papua: Liki, Bepondi, and Miossu (A. *Chaetodon semeion*, B. *Chaetodon ephippium*, C. *Chaetodon ornatissimus*, D. *Chaetodon rafflesi*, E. *Chaetodon meyeri*, F. *Chaetodon bennetti*, G. *Chaetodon ulietensis*, H. *Chaetodon ocellicaudus*, I. *Heniochus varius* (Photos by: Petrus Christianus Makatipu and Risandi Dwirama Putra).

2. The Kruskal Wallis test for Chaetodontidae Abundance

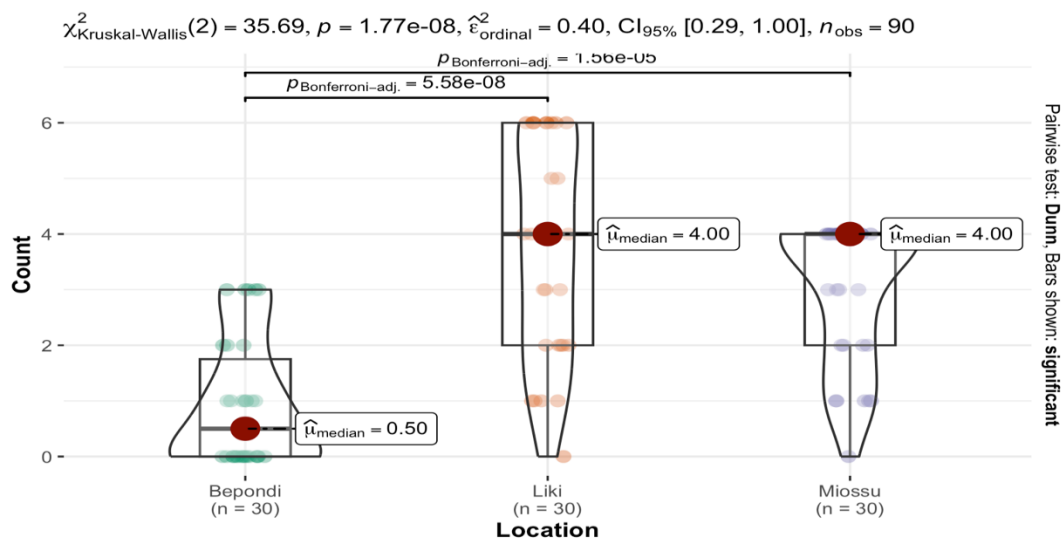


Fig. 3. Kruskal Wallis test for Chaetodontidae abundance.

The Kruskal-Wallis test results indicate significant differences in butterflyfish (Chaetodontidae) abundance among the three small outer island locations ($\chi^2 = 35.69$, $p = 1.77\text{e-}08$), with a large effect size ($\varepsilon^2_{\text{ordinal}} = 0.40$). Post hoc pairwise comparisons using Dunn's test with Bonferroni correction reveal that butterflyfish (Chaetodontidae) species abundance in Liki and Miossu islands is significantly higher than in Bepondi Island ($p = 5.58\text{e-}08$). The median abundance of butterflyfish (Chaetodontidae) was markedly lower at Bepondi Island (0.50) compared to both Liki and Miossu islands (4.00 for each), indicating a substantial disparity in reef fish assemblage structure. This pattern suggests that underlying environmental or ecological constraints such as different habitat complexity, coastal topography, or altered hydrodynamic conditions may be inhibiting both species richness and population abundance at Bepondi Island. These findings highlight the potential role of local habitat quality in shaping reef fish community composition within outer island ecosystems. The distribution spread in the violin plot suggests variability in species abundance across sites, particularly in Liki and Miossu. These findings emphasize the importance of localized habitat conditions influencing reef fish distribution and highlight Bepondi as a potential area of concern for conservation efforts.

3. The Correlation of Chaetodontidae Abundance Across Location

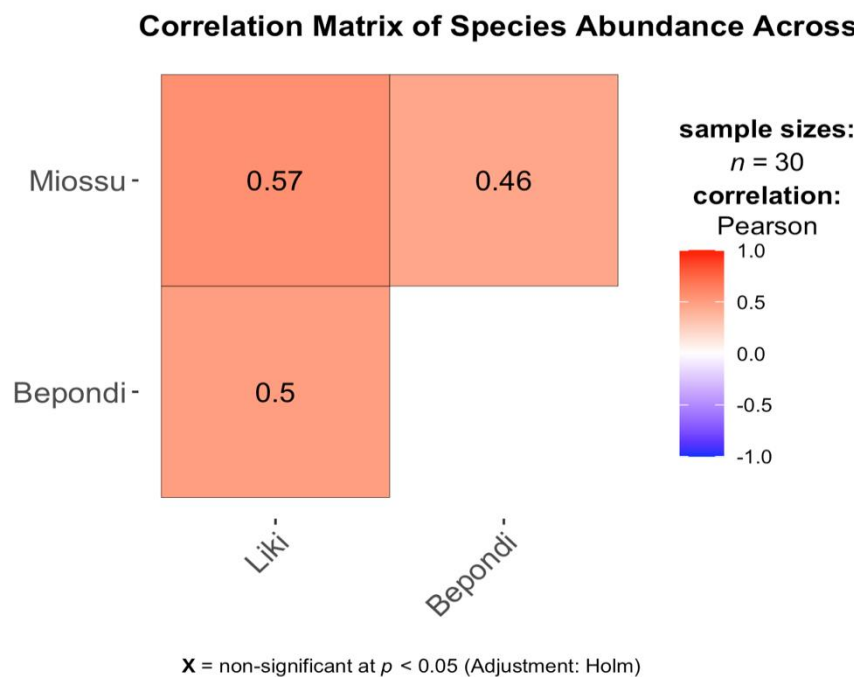


Fig. 4. The correlation of Chaetodontidae abundance across location (Liki, Bepondi and Miossu).

The correlation matrix of species abundance across Liki, Bepondi, and Miossu islands indicates moderate positive correlations between all pairs of locations, with the strongest correlation observed between Liki and Miossu ($r = 0.57$), followed by Liki and Bepondi ($r = 0.50$), and the weakest between Bepondi and Miossu ($r = 0.46$) (Fig. 4). These positive correlations suggest that the islands share ecological or environmental factors influencing Chaetodontidae species distributions among these islands. Despite being geographically distant, the similarity in butterflyfish assemblages between Liki and Miossu is higher than between either of them and the geographically closer Bepondi Island. This pattern likely reflects ecological and geomorphological differences rather than proximity alone. Specifically, Bepondi is characterized as a low island dominated by reef flats, whereas both Liki and Miossu are high islands with more structurally complex reef habitats. This shared topographic complexity between Liki and Miossu may support similar coral assemblages and benthic structures, which are critical determinants of Chaetodontidae diversity, given their strong reliance on coral cover for feeding and shelter. Additionally, the observed pattern may be influenced by variation in larval dispersal and recruitment dynamics. The connectivity between Liki and Miossu may be enhanced by favorable oceanographic currents that promote larval exchange over long distances, especially for species with longer pelagic larval durations. In contrast, Bepondi's different physical, geomorphology and ecological conditions may limit larval settlement from or to Liki and Miossu, despite being spatially closer. Collectively, these findings underscore the ecological connectivity among reef systems within the region.

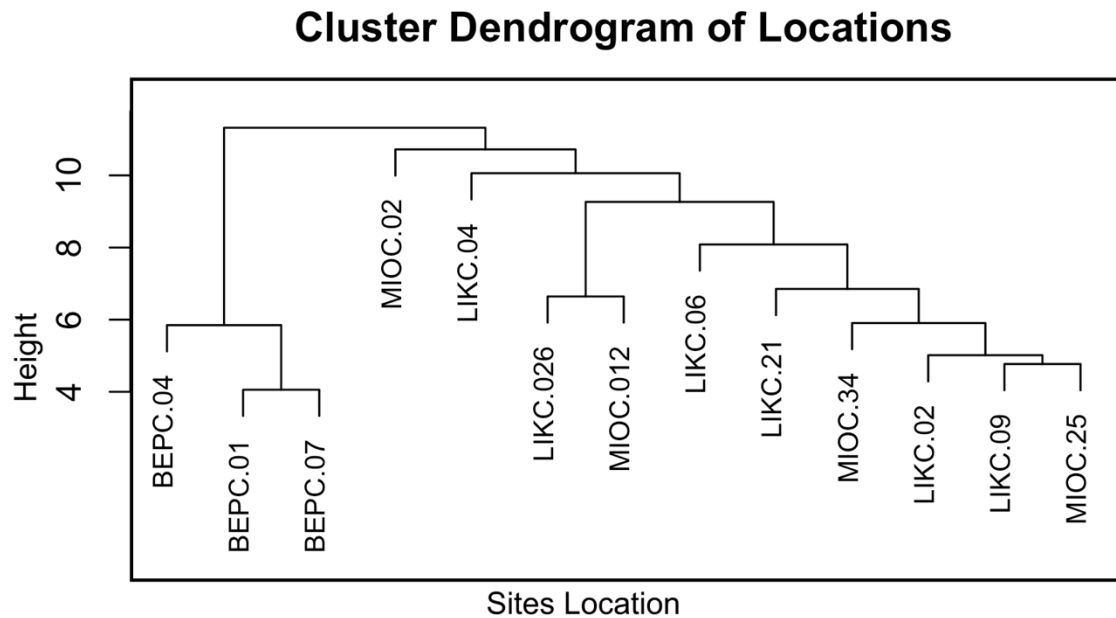


Fig. 5. The hierarchical clustering of sampling sites based on the similarity of Chaetodontidae species composition and abundance.

The cluster dendrogram provides a detailed hierarchical representation of the similarity in Chaetodontidae species composition and abundance across sampling sites on Bepondi Island (BEPC), the Miossu Islands (MIOC), and Liki Island (LIKC). The dendrogram groups sites based on the degree of similarity in their species assemblages, with branch lengths indicating the level of dissimilarity; shorter branches denote higher similarity, while longer branches reflect greater differences. Sites from Bepondi Island (BEPC.04, BEPC.01, BEPC.07) form a distinct cluster suggest that these locations share similar environmental conditions and species compositions, likely due to the island's flat and homogeneous coral reef landscape. This homogeneity may limit habitat diversity, resulting in fewer ecological niches and lower species variation compared to the other islands. In contrast, sites along the Miossu Islands (MIOC.02, MIOC.012, MIOC.25, MIOC.34) and Liki Island (LIKC.04, LIKC.026, LIKC.06, LIKC.21, LIKC.02, LIKC.09) are grouped into clusters that reflect their more complex and varied coral reef ecosystems. The hilly topography of these islands likely provides diverse habitats, creating multiple ecological niches that support a wider range of Chaetodontidae species. This is evident in the clustering of sites like MIOC.02 and LIKC.04, which share similar species assemblages due to their comparable habitat complexity. The height axis (ranging from 4 to 10) quantifies the level of dissimilarity between clusters, with higher values indicating more pronounced differences. For instance, the greater dissimilarity between BEPC clusters and MIOC/LIKC clusters underscores the significant impact of island type and habitat complexity on species distribution. This dendrogram highlights the critical role of environmental heterogeneity in shaping Chaetodontidae community structure, emphasizing the importance of habitat diversity in supporting species richness and ecological resilience. These findings provide valuable insights for conservation strategies aimed at preserving reef fish biodiversity in the region.

DISCUSSION

Our findings indicate that the three outer islands of eastern Indonesia Liki, Bepondi, and Miossu exhibit high reef fish diversity, with 30 species of butterflyfish (Chaetodontidae) recorded (Table 2). Several ecological and biogeographic factors likely contribute to the high diversity of butterflyfish (Chaetodontidae) observed in the outer islands of West Papua, particularly around Liki, Bepondi, and Miossu. Geographically, these islands are situated near the Raja Ampat archipelago, within the core of the Coral Triangle, which is widely recognized as the global epicenter of tropical marine biodiversity (*Zhao et al., 2025*). This proximity may exert a strong regional influence on species richness through larval connectivity, oceanographic transport, and ecological spillover from nearby biodiversity hotspots. Previous studies have documented a clear gradient of increasing Chaetodontidae richness from western to eastern Indonesia, corresponding with proximity to Raja Ampat. For example, only two species were recorded in Bintan, western

Indonesia (**Ardian *et al.*, 2020**). While, nine species were detected in the Thousand Islands (**Maduppa *et al.*, 2013**), and 21 species in the Spermonde Archipelago, central Indonesia (**Putra *et al.*, 2024**). This number peaked to 27 species in the Dampier Strait, Raja Ampat (**Maduppa *et al.*, 2018**). These findings support the hypothesis that the species richness of butterflyfish species in Liki, Bepondi, and Miossu is shaped by their location within a broader biogeographic transition zone toward the highest global center of reef fish diversity.

Another contributing factor to the high species richness of butterflyfish (Chaetodontidae) observed in Liki, Bepondi, and Miossu islands is their relative remoteness from the mainland, which provides a natural buffer against widespread anthropogenic disturbances such as coastal development, pollution, and overfishing (Fig. 1). Previous studies have highlighted that coral reef biodiversity is increasingly at risk due to anthropogenic disturbances (**Maduppa *et al.*, 2013**; **Holbrook *et al.*, 2015**). Remarkably, the frequency and intensity of such pressures escalated over the past century (**Wilson *et al.*, 2006**). The influence of anthropogenic pressures on butterflyfish (Chaetodontidae) assemblages has been well-documented in previous studies, underscoring the sensitivity of this taxonomic group to human-induced reef degradation. For instance, **Rina *et al.* (2020)** reported that high tourism intensity at Samber Gelap Island significantly altered the composition and abundance of Chaetodontidae species, likely due to habitat disturbance and increased stress on coral communities. Similarly, (**Hamuna *et al.*, 2019**) found that coral reefs along the coastline of Jayapura City were classified as severely degraded, primarily as a result of coastal development, pollution, and other anthropogenic disturbances. In these areas, only nine Chaetodontidae species were recorded, highlighting a potential link between habitat degradation and reduced reef fish diversity. These findings contrast with the high species richness observed at the relatively undisturbed Liki, Bepondi, and Miossu islands, further supporting the role of low human impact and traditional ecological stewardship in maintaining reef fish biodiversity.

These disturbances, when acting simultaneously, significantly alter diversity components (**González-Sansón & Aguilar, 2010**), leading to the progressive decline in reef health and productivity observed in many small islands (**Raymundo *et al.*, 2019**). Localized anthropogenic threats in small island reefs pose a serious risk of irreversible biodiversity loss, ultimately affecting the essential ecosystem services they provide (**Hughes *et al.*, 2003**; **Sala & Knowlton, 2006**; **Worm *et al.*, 2006**; **Wild *et al.*, 2011**; **Rogers, 2013**). The increasing intensity of these stressors is often linked to rising human populations (**Baum *et al.*, 2015**), which drive higher levels of marine resource exploitation and contribute to the degradation of coastal ecosystems (**D'Agata *et al.*, 2014**; **Graham, 2014**; **Duffy *et al.*, 2016**), impacting functional group diversity (**Nystrom *et al.*, 2000**).

Although two of the three islands Liki and Bepondi are inhabited, their communities are composed of traditional, locally rooted populations who practice customary marine tenure systems. Notably, the application of Sasi, an indigenous resource management

practice that regulates seasonal access and harvest restrictions, appears to play a critical role in maintaining the ecological integrity of local reef systems. Field observations support the notion that such customary governance contributes to the preservation of marine biodiversity, including the conservation of ecologically sensitive indicator taxa such as Chaetodontidae. Sasi system is supported by local communities whose livelihoods depend on marine resources. These communities predominantly employ traditional, environmentally friendly fishing practices, which play a crucial role in maintaining the sustainability of marine ecosystems. Such fishing methods help preserve keystone species, thereby promoting ecosystem productivity and stability. The continued use of these sustainable practices reinforces the ecological resilience of these reef systems. One of the most effective community-based conservation efforts in Papua that has positively impacted Chaetodontidae populations is the traditional practice of Sasi. Sasi plays a crucial role in local marine conservation, as it involves community participation in regulating resource use and protecting marine biodiversity (**Jurnalica et al., 2008; Febyarandika & Chafid, 2016; Sumarsono & Wasa, 2019; Putri et al., 2020**). Several studies have demonstrated that community-led management of small-scale marine protected areas (MPAs), where fishing is restricted, has resulted in increased abundance of Chaetodontidae species (**Kulbicki et al., 2013**). Similarly, a study in Fiji found that community-based management of marine protected areas led to a significant increase in reef fish biomass (**Golden et al., 2014; Albert et al., 2016**). Researchers attribute the success of these conservation efforts largely to the presence of Sasi and the active involvement of local communities in decision-making and the implementation of environmental conservation measures. We also highlight the abundance of reef fish might also be affected by coral cover or reef complexity, especially the butterflyfish which are regarded as reef health indicator (**Nugraha et al., 2020**)

The observed diversity and abundance of Chaetodontidae species across Liki, Bepondi, and Miossu islands may be influenced by a combination of factors, including geographic remoteness, limited anthropogenic disturbance, oceanographic conditions, and underlying geomorphological characteristics of each island. Beyond their isolation, oceanographic variables such as sea surface temperature, salinity, current regimes, and nutrient fluxes are known to significantly shape coral reef fish assemblages, particularly for coral-dependent taxa like Chaetodontidae. These environmental parameters directly affect coral reef health, food availability, larval dispersal, and habitat complexity each of which plays a critical role in supporting diverse and stable butterflyfish communities. Additionally, the distinct island geomorphologies ranging from steep, high island slopes to broad, shallow reef flats further modulate habitat structure, influencing the spatial distribution and ecological niches available to butterflyfish species (**Schoepf et al., 2023; Isdianto et al., 2024; Nhat et al., 2024**). These species are highly dependent on coral reef habitats (**Hoegh-Guldberg et al., 2019; Zhou et al. 2024**). Variations in these oceanographic parameters can impact coral health, influencing the distribution and

abundance of butterflyfish species (Pratchett & Berumen, 2008; MacDonald *et al.*, 2021). Strong currents and upwelling events can enhance nutrient cycling, supporting coral growth and increasing fish diversity (D'angelo & Wiedenmann, 2014), while changes in temperature and salinity may induce coral stress (Dias *et al.*, 2019; Ding *et al.*, 2022), potentially reducing habitat suitability for Chaetodontidae.

The intricate relationship between environmental factors and biological interactions underscores the complexity of reef fish community structure, particularly within the Chaetodontidae family in the outer islands of Papua. This highlights the necessity for sustained oceanographic monitoring to better understand long-term population dynamics. Our findings reveal that island type significantly shapes Chaetodontidae species diversity, abundance, and distribution. For example, Bepondi Island exhibits lower Chaetodontidae diversity and abundance compared to Liki and Miossu islands (Fig.3), likely due to its reef flat and shallow topography condition. In contrast, the reef slope topography and more expansive landscapes coral ecosystem of Liki and Miossu provide a greater variety of habitats, fostering higher species diversity. These landscape-driven variations in habitat complexity play a critical role in influencing Chaetodontidae assemblages, as different island types offer distinct ecological niches (Luo *et al.*, 2023). This study emphasizes the importance of considering geomorphological characteristics in conservation and monitoring efforts to preserve reef fish biodiversity in these regions. Islands characterized by more diverse coral reef ecosystems, such as those with hilly topographies, provide heterogeneous habitats that foster the creation of multiple ecological niches, enabling Chaetodontidae species to thrive. These varied niches support greater species variation and higher diversity, as observed in islands like Liki and Miossu. In contrast, islands with more homogeneous, flat, and shallow coral reef landscapes, such as Bepondi Island, offer fewer ecological niches, potentially limiting habitat diversity for Chaetodontidae. Consequently, the overall diversity of Chaetodontidae on Bepondi is lower compared to the structurally complex environments of Liki and Miossu (Figs. 4, 5). These findings underscore the critical role of island type in shaping ecological diversity, with niche availability emerging as a key determinant in sustaining a rich array of species. This emphasizes the importance of considering habitat complexity in conservation strategies aimed at preserving reef fish biodiversity.

CONCLUSION

This study recorded a total of 30 butterflyfish (Chaetodontidae) species across Liki, Bepondi, and Miossu islands, with significantly higher fish density observed in Liki and Miossu compared to Bepondi. The presence of one Near Threatened (*Chaetodon trifascialis*) species highlights the importance of continuous monitoring and conservation efforts. Statistical analysis using the Kruskal-Wallis test confirmed significant differences in species abundance, with Liki and Miossu supporting greater biodiversity than Bepondi. These findings underscore the ecological significance of these remote islands and the need

for effective management strategies to maintain reef fish populations and overall coral reef health.

ACKNOWLEDGMENT

We express our sincere gratitude to the Coral Reef Rehabilitation Management Program (COREMAP) and the Research Centre for Oceanography, Indonesian Institute of Sciences (PRO-LIPI), for their financial and logistical support in making the Nusa Manggala Expedition a success. We also extend our appreciation to the Indonesian government and all scientists involved in the expedition. Special thanks go to the captain and crew of Baruna Jaya VIII for their hospitality throughout the research. We acknowledge Mr. Udhie E. Hernawan for his role as project coordinator and Mr. Bayu Prayudha for his contributions to GIS analysis. Additionally, we are grateful to Mr. Frenslly Damnianus Hukom, Miss Sasanti, Mr. Muin Sinaga, Mr. Karnan, Mr. Acel Arafat, Mr. Jemi Ferizal, Mr. Kunto Wiobowo, Mr. Nagib Isa Idrus, and Mr. Yunaldi for their invaluable assistance in identifying and verifying fish species recorded in this study.

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